

# Siddhartha Reddy Munugala

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## SUMMARY

Founding engineer and AI-first builder who has shipped two live platforms with 1,321 combined registered users. Experienced in building community platforms, submission pipelines, data-driven recommendation systems, and AI-powered analytics tools from scratch. Currently leading zero-to-one development of EKO, an AI audio platform in active build. Background in customer discovery (50+ user interviews), startup operations, and rapid product iteration -- bringing a founder mindset and data-driven rigor to every problem.

## SKILLS

AI / ML: LLMs, RAG, prompt engineering, agentic workflows, XGBoost, SHAP, PyTorch, inference optimization, Claude API, Gemini API, OpenAI API, LangChain

Languages: Python, TypeScript, JavaScript, SQL

Frameworks: Next.js 14, React, Node.js, FastAPI, Tailwind CSS

Data & Infra: PostgreSQL, Neon Postgres, Supabase, Redis, Docker, Vercel, Render, GitHub Actions, CI/CD, HuggingFace Inference API

Other: WebRTC, WebSockets, REST APIs, NetworkX, tree-sitter, Google Apps Script, distributed systems

## EXPERIENCE

### Co-Founder and Lead Engineer — VibeIn LLC

Jan 2025 - Present

vibein.us / arenahq.us

- Built ArenaHQ (arenahq.us) -- a live competitive platform with ELO ranking, user submission pipelines, and peer recommendation matrices serving 800+ registered users; architecture directly analogous to a venture scout submission and founder-ranking system.
- Built VibeIn (vibein.us) -- an AI-powered community matching platform using data-driven peer recommendations; 521 registered users across a single early community vertical, with 10 percent active retention before scaling to additional communities.
- Diagnosed and resolved 8+ production-critical failures maintaining 99 percent or higher uptime; owned full-stack development, cloud deployment, and product iteration across both platforms simultaneously.
- Conducted 50+ structured user interviews driving product iteration; managed end-to-end Vercel deployment pipelines and zero-downtime release cadences.

### IT Coordinator and Publicity Chair — NSLS Executive Board

Sep 2024 - Present

University of South Dakota

- Built Python and Google Apps Script data pipelines for member tracking, recruitment analytics, and communication automation -- reducing onboarding time by 50 percent (4 weeks to 2 weeks); managed infrastructure across 700+ campuses.

### Production Operations Staff — USD Athletics

Aug 2025 - Present

University of South Dakota

- Execute timing-critical live broadcast operations under zero-error constraints -- demonstrating discipline and execution under pressure.

## PROJECTS

### PRISM — AI-Powered Deal Flow and Risk Intelligence System

2026

- Built an ML pipeline analyzing 45 signals -- code complexity, dependency graphs, contributor patterns, churn metrics -- to score and rank engineering risk; XGBoost + SHAP explainability, AUC 0.83. Architecture directly translatable to AI-driven startup opportunity scoring and founder signal analysis for a VC platform.

### EKO — AI Audio Platform (Active Zero-to-One Build)

2026 - Present

- Leading end-to-end development of a three-mode AI audio platform: Studio (vocal sketch to BGM variations via Google Lyria 3), BGM Dub (scene-matched music for video), and Voice Dub (multilingual speech replacement via Demucs). Resolved critical infrastructure constraints (78 percent memory footprint reduction) pre-launch. Stack: Next.js 14, FastAPI, Neon Postgres, Gemini Flash, Demucs, librosa, faster-whisper.

### PETRI-Verse — 62-Agent Autonomous AI Civilization Simulation

2025

- Concurrent 62-agent LLM simulation with tick-based event loop across distributed async agents; produced a novel AI safety finding around emergent self\_modify() behavior with implications for multi-agent containment research.

### ALASKA — Encrypted AI Assistant

2026

Top 10, Wildfire Labs Hackathon

- AES-256-GCM encrypted LLM assistant with VA routing algorithm, crisis detection, and accessibility modes -- built in 48 hours; Top 10 finish demonstrating rapid execution under competitive conditions.

### WTFcode — Natural Language Agentic Developer CLI

2026

- Multi-agent CLI converting natural language to structured multi-step code execution pipelines; reduced token overhead by 90 percent via importmap compression.

## EDUCATION

### B.S. Computer Science, Minor in Innovation and Entrepreneurship

Expected May 2027

University of South Dakota

Coursework: Data Structures, Algorithms, Systems Programming, Database Systems, Software Engineering

NSF I-Corps Customer Discovery Training • NSLS National Leadership Inductee

## LEADERSHIP

- AI and Entrepreneurship Workshop Facilitator (Self-Initiated) -- Designed and led community sessions for teenagers ages 13-16 on AI, product building, customer discovery, and startup fundamentals.
- USD INFORMS | NSLS Executive Board | USD Marketing Club -- operational leadership and cross-functional execution across multiple organizations simultaneously.